

Practice Assessment

A low-stakes rehearsal | about 30 minutes | Solutions included for self-checking

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

Try before reading the solutions

1. Convert BED[4,9) to one-based inclusive coordinates and give the length.

2. Three independent cultures each yield two aliquots and 1000reads per aliquot. What is the biological replicate count?

3. For p-values 0.01 and 0.08, calculate BH-adjusted values and choose discoveries at 0.05.

4. Why might a random row split be unsuitable for repeated samples from the same donor?

5. A structure has confident local domains but uncertain relative placement. Which confidence view helps reveal the second issue?

6. A model output says a VUS proves a diagnosis. Write a safer sentence.

Self-check explanations

1

[5,9], length 5. Only the start changes coordinate convention. [S03,S04]

2

Three cultures. Six aliquots are technical subdivisions; reads are observations from measurement, not six or 6000 independent cultures. [S10]

3

Adjusted 0.02 and 0.08; only the first passes. This is a two-test family, not a gene-specific probability of truth. [S08]

4

The same donor can occur in train and test; the evaluation can exploit donor signatures. Hold out donors to evaluate new donors. [S11]

5

PAE can reveal relative-placement uncertainty; high local pLDDT does not settle domain orientation or binding. [S16]

6

The variant remains of uncertain significance in the stated context; current evidence does not establish causality. Qualified contextual assessment and additional evidence may be needed. [S18,S19]

For every answer ask: what was measured, what was assumed, what remains uncertain? This practice set is not the confidential midterm or final and does not predict every examination item.